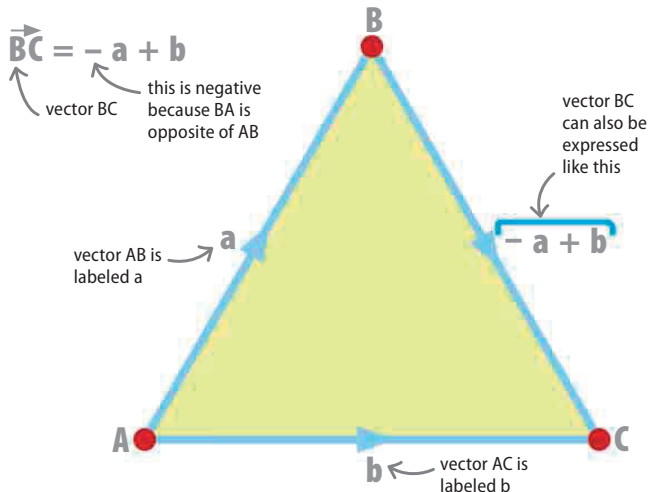


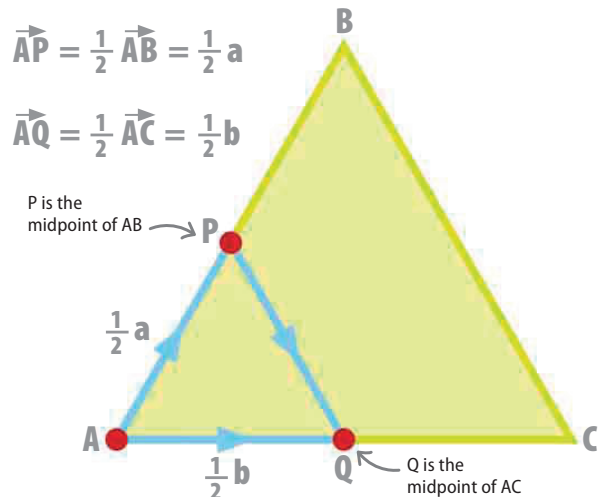
## Working with vectors in geometry

Vectors can be used to prove results in geometry. In this example, vectors are used to prove that the line joining the midpoints of any two sides of a triangle is parallel to the third side of the triangle, as well as being half its length.

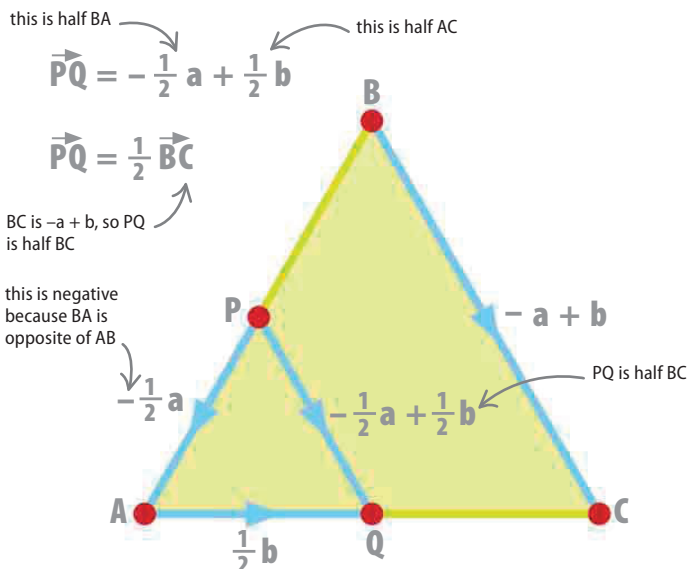
**First, choose 2 sides** of triangle ABC, in this example AB and AC. Mark these sides as the vectors  $\mathbf{a}$  and  $\mathbf{b}$ . To get from B to C, go along BA and then AC, rather than BC. BA is the vector  $-\mathbf{a}$  because it is the opposite of AB, and AC is already known ( $\mathbf{b}$ ). This means vector BC is  $-\mathbf{a} + \mathbf{b}$ .



**Second, find the midpoints** of the two sides that have been chosen (AB and AC). Mark the midpoint of AB as P, and the midpoint of AC as Q. This creates three new vectors: AP, AQ, and PQ. AP is half the length of vector  $\mathbf{a}$ , and AQ is half the length of vector  $\mathbf{b}$ .



**Third, use the vectors**  $\frac{1}{2}\mathbf{a}$  and  $\frac{1}{2}\mathbf{b}$  to find the length of vector PQ. To get from P to Q go along PA then AQ. PA is the vector  $-\frac{1}{2}\mathbf{a}$  because it is the opposite of AP, and AQ is already known to be  $\frac{1}{2}\mathbf{b}$ . This means vector PQ is  $-\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$ .



**Fourth, make the proof.** The vectors PQ and BC are in the same direction and are therefore parallel to each other, so the line PQ (which joins the midpoints of the sides AB and AC) must be parallel to the line BC. Also, vector PQ is half the length of vector BC, so the line PQ must be half the length of the line BC.

